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Chapter Review

Problems: Additional Problems

37. *Why is the following situation impossible?* In the Bohr model of the hydrogen atom, an electron moves in a circular orbit about a proton. The model states that the electron can exist only in certain allowed orbits around the proton: those whose radius r satisfies $r = n^2(0.052\ 9\ \text{nm})$, where $n = 1, 2, 3, \ldots$. For one of the possible allowed states of the atom, the electric potential energy of the system is $-13.6\ \text{eV}$.

38. On a dry winter day, you scuff your leather-soled shoes across a carpet and get a shock when you extend the tip of one finger toward a metal doorknob. In a dark room, you observe a spark perhaps 5 mm long. Make order-of-magnitude estimates of

    (a) your electric potential and

    (b) the charge on your body before you touch the doorknob. Explain your reasoning.

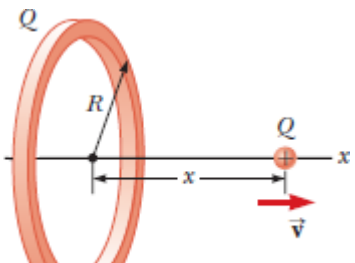
39. QC S

    (a) Use the exact result from Example 24.4 to find the electric potential created by the dipole described in the example at the point $(3a, 0)$.

    (b) Explain how this answer compares with the result of the approximate expression that is valid when x is much greater than a.

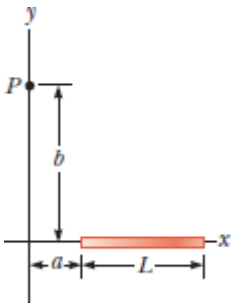
40. *Why is the following situation impossible?* You set up an apparatus in your laboratory as follows. The x axis is the symmetry axis of a stationary, uniformly charged ring of radius $R = 0.500\ \text{m}$ and charge $Q = 50.0\ \mu\text{C}$ (Fig. P24.40). You place a particle with charge $Q = 50.0\ \mu\text{C}$ and mass $m = 0.100\ \text{kg}$ at the center of the ring and arrange for it to be constrained to move only along the x axis. When it is displaced slightly, the particle is repelled by the ring and accelerates along the x axis. The particle moves faster than you expected and strikes the opposite wall of your laboratory at 40.0 m/s.

Figure P24.40



41. S The thin, uniformly charged rod shown in Figure P24.41 has a linear charge density λ. Find an expression for the electric potential at P.

Figure P24.41



42. S A Geiger–Mueller tube is a radiation detector that consists of a closed, hollow, metal cylinder (the cathode) of inner radius r_a and a coaxial cylindrical wire (the anode) of radius r_b (Fig. P24.42a). The charge per unit length on the anode is λ, and the charge per unit length on the cathode is $-\lambda$. A gas fills the space between the electrodes. When the tube is in use (for example, in measuring radioactivity from fruit in Fig. P24.42b) and a high-energy elementary particle passes through this space, it can ionize an atom of the gas. The strong electric field makes the resulting ion and electron accelerate in opposite directions. They strike other molecules of the gas to ionize them, producing an avalanche of electrical discharge. The pulse of electric current between the wire and the cylinder is counted by an external circuit.

(a) Show that the magnitude of the electric potential difference between the wire and the cylinder is

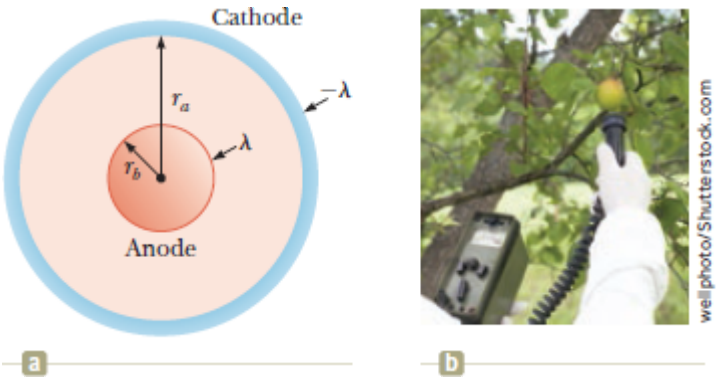
$$\Delta V = 2k_e\lambda \ln\left(\frac{r_a}{r_b}\right)$$

(b) Show that the magnitude of the electric field in the space between cathode and anode is

$$E = \frac{\Delta V}{\ln\left(r_a/r_b\right)}\left(\frac{1}{r}\right)$$

where r is the distance from the axis of the anode to the point where the field is to be calculated.

Figure P24.42



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43. **Review.** Two parallel plates having charges of equal magnitude but opposite sign are separated by 12.0 cm. Each plate has a surface charge density of $36.0\ \mathrm{nC/m^2}$. A proton is released from rest at the positive plate. Determine

(a) the magnitude of the electric field between the plates from the charge density,

(b) the potential difference between the plates,

(c) the kinetic energy of the proton when it reaches the negative plate,

(d) the speed of the proton just before it strikes the negative plate,

(e) the acceleration of the proton, and

(f) the force on the proton.

(g) From the force, find the magnitude of the electric field.

(h) How does your value of the electric field compare with that found in part (a)?

44. S When an uncharged conducting sphere of radius a is placed at the origin of an xyz coordinate system that lies in an initially uniform electric field $\vec{\mathbf{E}} = E_0\hat{\mathbf{k}}$, the resulting electric potential is $V(x, y, z) = V_0$ for points inside the sphere and

$$V(x, y, z) = V_0 - E_0 z + \frac{E_0 a^3 z}{(x^2 + y^2 + z^2)^{3/2}}$$

for points outside the sphere, where V_0 is the (constant) electric potential on the conductor. Use this equation to determine the x, y, and z components of the resulting electric field

(a) inside the sphere and

(b) outside the sphere.

45. GP S A solid, insulating sphere of radius a has a uniform charge density throughout its volume and a total charge Q. Concentric with this sphere is an uncharged, conducting, hollow sphere whose inner and outer radii are b and c as shown in Figure P24.45. We wish to understand completely the charges and electric fields at all locations.

(a) Find the charge contained within a sphere of radius $r < a$.

(b) From this value, find the magnitude of the electric field for $r < a$.

(c) What charge is contained within a sphere of radius r when $a < r < b$?

(d) From this value, find the magnitude of the electric field for r when $a < r < b$.

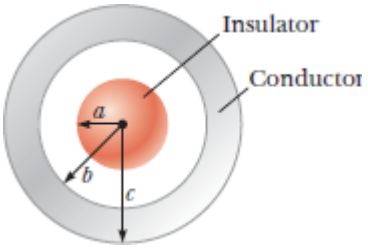
(e) Now consider r when $b < r < c$. What is the magnitude of the electric field for this range of values of r?

(f) From this value, what must be the charge on the inner surface of the hollow sphere?

(g) From part (f), what must be the charge on the outer surface of the hollow sphere?

(h) Consider the three spherical surfaces of radii a, b, and c. Which of these surfaces has the largest magnitude of surface charge density?

Figure P24.45



46. QIC A hollow, metallic, spherical shell has exterior radius 0.750 m, carries no net charge, and is supported on an insulating stand. The electric field everywhere just outside its surface is 890 N/C radially toward the center of the sphere. Explain what you can conclude about

(a) the amount of charge on the exterior surface of the sphere and the distribution of this charge,

(b) the amount of charge on the interior surface of the sphere and its distribution, and

(c) the amount of charge inside the shell and its distribution.

47. QIC For the configuration shown in Figure P24.45, suppose $a = 5.00\ \text{cm}$, $b = 20.0\ \text{cm}$, and $c = 25.0\ \text{cm}$. Furthermore, suppose the electric field at a point 10.0 cm from the center is measured to be $3.60 \times 10^3\ \text{N/C}$ radially inward and the electric field at a point 50.0 cm from the center is of magnitude 200 N/C and points radially outward. From this information, find

(a) the charge on the insulating sphere,

(b) the net charge on the hollow conducting sphere,

(c) the charge on the inner surface of the hollow conducting sphere, and

(d) the charge on the outer surface of the hollow conducting sphere.

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